

Brando Vasquez Rodriguez

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AVAILABLE 2nd CO-OP SPRING 2027 | 3rd CO-OP FALL 2027

EDUCATION

Bachelor of Science, Electrical Engineering

University of Cincinnati, Cincinnati, OH

Graduating: May 2029

GPA: 3.97

SKILLS

Software: Revit, AutoCAD, MATLAB, C++, Python, SQL, Verilog, LabVIEW, Arduino IDE, LTspice, Autodesk Fusion 360, Microsoft Office 365.

Hardware/Lab: Oscilloscope, Function Generator, Multimeter, Soldering, Embedded Systems, Sensor Integration.

Languages: English, Spanish.

PROFESSIONAL EXPERIENCE

Electrical Engineering Co-op

June 2026 - Present

IMEG, Blue Ash, OH

- Supported Revit and AutoCAD electrical document updates for power plans, lighting plans, and single-line diagrams.
- Supported existing-conditions site observations by documenting electrical equipment for renovation drawing updates.
- Assisted with single-line diagram and AGi32 calculations, checking transformer/panel data and AutoCAD backgrounds.

Undergraduate Research Assistant

October 2025 - Present

R.O.S.E Lab, University of Cincinnati

- Analyzed time-series physiological data using statistical methods to examine stress patterns in engineering students.
- Performed structured literature reviews and data organization to support experimental design and research methodology.
- Collaborated with faculty and interdisciplinary teams to maintain research documentation and data reproducibility.

PROJECTS & ADDITIONAL EXPERIENCE

Gym WRO - Wearable Rep Optimizer (Project)

May 2026 - Present

University of Cincinnati, Cincinnati, OH

- Developing a wearable prototype using muscle-sensing and motion data to track rep quality and fatigue trends.
- Designed system architecture to integrate EMG/IMU sensing, ESP32 control, battery power, and real-time LED feedback.
- Planned prototype testing for biceps curls to evaluate rep tracking, range of motion, fatigue trends, and user feedback.

Light-Responsive Alarm System (ESP32 Project)

March 2026 - May 2026

University of Cincinnati, Cincinnati, OH

- Designed an ESP32-based embedded system with sensors, LCD display, and buzzer for responsive monitoring.
- Processed analog signals (12-bit ADC) and applied averaging to reduce noise and stabilize sensor measurements.
- Implemented Wi-Fi time synchronization and designed a custom CAD enclosure in Fusion 360 for the prototype system.

Student Ambassador

August 2025 - May 2026

University of Cincinnati Blue Ash, Blue Ash, OH

- Represented UCBA as a campus leader by serving as the first point of contact for over 4,800 students and their families.
- Conducted campus tours and orientations for approximately 1,800+ prospective students and guests.
- Served as a Student Life liaison, coordinating events and digital outreach with staff to enhance student engagement.

Python & SQL Trainee (Project-Based)

May 2025 - August 2025

The Global Career Accelerator, Remote

- Analyzed industry datasets for Intel and The Grammys to evaluate hardware sustainability and platform user engagement.
- Applied Python (Pandas, NumPy) and SQL to clean, transform, and validate structured datasets in Jupyter Notebook.
- Created data visualizations and summarized findings into clear, actionable recommendations for stakeholders.

Autonomous Parking Robot (Project)

January 2025 - April 2025

University of Cincinnati, Cincinnati, OH

- Designed and built an autonomous LEGO EV3 vehicle using iterative engineering design and testing.
- Programmed multi-sensor control (color, gyro, ultrasonic, touch) in LabVIEW for autonomous parking.

LEADERSHIP, ACTIVITIES, & SERVICE

STEMulates Robotics Volunteer

December 2025 - January 2026

The Gaskins Foundation, Cincinnati, OH

- Guided K-12 students through small hands-on robotics projects to introduce engineering problem-solving.

Vice President & Event Coordinator

October 2024 - May 2025

Latinos en Acción (LEA) Affinity Club, University of Cincinnati Blue Ash

- Led planning and execution of campus events, coordinating logistics, resources, and outreach to strengthen engagement.